

Appendix for An End-to-End Cost-Focused Wind Power Forecasting Framework Based on Convex Hulls

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1 Appendix

1.1 Relationship Between Forecasting and Scheduling

Let Ξ_t denote the set of decision variables at time t , including wind power output $P_{i,t}^{WT}$, distributed generation $P_{i,t}^{DG}$, and grid power P_t^{MG} . Let y_t be the ground truth wind power, and $\hat{y}_t$ (denoted as $\bar{P}_{i,t}^{WT}$ in the optimization model) be the predicted wind power availability.

The standard statistical loss function aims to minimize the Euclidean distance between the prediction and the ground truth:

$$\mathcal{L}_{MSE}(\hat{y}_t, y_t) = \frac{1}{2} \|\hat{y}_t - y_t\|^2 \quad (1)$$

The gradient descent direction with respect to the prediction $\hat{y}_t$ is determined by:

$$\nabla_{\hat{y}_t} \mathcal{L}_{MSE} = \hat{y}_t - y_t \quad (2)$$

Following the formulation in Eq.(2) and constraints (1a)-(1f), the day-ahead scheduling problem can be viewed as a parametric optimization problem where the minimal cost $J(\hat{y}_t)$ is a function of the forecast parameter $\hat{y}_t$:

$$J(\hat{y}_t) = \min_{\Xi_t} C^{obj}(\Xi_t, \hat{y}_t) = \min_{\Xi_t} \sum_{t=1}^T \left[\pi_t^{MG} P_t^{MG} + \sum_{i=1}^N (\pi_i^{DG} P_{i,t}^{DG} + \pi_t^{cur} (\hat{y}_t - P_{i,t}^{WT})) \right] \quad (3)$$

Subject to the forecast availability constraint and system physical constraints:

$$P_{i,t}^{WT} - \hat{y}_t \leq 0 \quad (4a)$$

$$\mathbf{h}(\Xi_t) \leq \mathbf{0} \quad (4b)$$

Gradient Structure of Economic Objective: The gradient of the scheduling cost $J(\hat{y}_t)$ with respect to the forecast parameter $\hat{y}_t$ is discontinuous and is determined by the activation status of the physical grid constraints.

Proof. We form the Lagrangian function $\mathcal{L}_{opt}$ for the optimization problem defined by (3)-(5). Let $\lambda_{i,t} \geq 0$ be the dual variable associated with the wind power availability constraint (4), and $\boldsymbol{\mu}_t$ be the vector of dual variables for physical constraints (5).

$$\mathcal{L}_{opt} = C^{obj}(\Xi_t, \hat{y}_t) + \sum_{i=1}^N \lambda_{i,t} (P_{i,t}^{WT} - \hat{y}_t) + \boldsymbol{\mu}_t^T \mathbf{h}(\Xi_t) \quad (5)$$

Substituting the explicit form of C^{obj} from Eq.(2) into (6):

$$\mathcal{L}_{opt} = [\dots + \pi_t^{cur}(\hat{y}_t - P_{i,t}^{WT})] + \lambda_{i,t}(P_{i,t}^{WT} - \hat{y}_t) + \dots \quad (6)$$

According to the Envelope Theorem, the gradient of the optimal cost function $J(\hat{y}_t)$ with respect to the parameter $\hat{y}_t$ is the partial derivative of the Lagrangian:

$$\nabla_{\hat{y}} J = \frac{\partial \mathcal{L}_{opt}}{\partial \hat{y}_t} = \pi_t^{cur} - \lambda_{i,t} \quad (7)$$

Based on the Karush-Kuhn-Tucker (KKT) complementarity slackness condition for constraint:

$$\lambda_{i,t} \cdot (P_{i,t}^{WT} - \hat{y}_t) = 0, \quad \lambda_{i,t} \geq 0 \quad (8)$$

Objective Mismatch: There exists a domain where the optimization direction of $\mathcal{L}_{MSE}$ contradicts the optimization direction of $J(\hat{y}_t)$.

Proof. Since $\hat{y}_t < y_t$, the gradient is negative, i.e.,

$$\nabla_{\hat{y}} \mathcal{L}_{MSE} < 0 \quad (9)$$

The optimization gradient will become positive at some stage:

$$\nabla_{\hat{y}} J \approx > 0 \quad (10)$$

The inner product of the two descent directions is negative:

$$\langle -\nabla \mathcal{L}_{MSE}, -\nabla J \rangle < 0 \quad (11)$$

1.2 Convex Hull

Let $J(\hat{y})$ denote the true minimal scheduling cost parameterized by the forecast value $\hat{y}$. Based on the optimization problem defined in Eq.(2), $J(\hat{y})$ is given by:

$$J(\hat{y}) = \inf_{\Xi \in \mathcal{F}} \{C^{obj}(\Xi, \hat{y}) \mid P^{WT}(\Xi) \leq \hat{y}\} \quad (12)$$

where $\mathcal{F}$ represents the feasible set defined by physical grid constraints (Eq.(1a)-(1f)). Due to the discrete activation of these constraints, $J(\hat{y})$ is inherently non-smooth, making direct gradient optimization prone to local minima.

The Fenchel conjugate $J^*(\nu)$ and biconjugate $J^{**}(\hat{y})$ are defined as:

$$J^*(\nu) = \sup_{\hat{y}} \{\langle \nu, \hat{y} \rangle - J(\hat{y})\}, \quad J^{**}(\hat{y}) = \sup_{\nu} \{\langle \nu, \hat{y} \rangle - J^*(\nu)\} \quad (13)$$

According to the Fenchel-Moreau theorem, $J^{**}(\hat{y})$ corresponds to the lower convex envelope of $J(\hat{y})$. It represents the tightest possible convex lower bound of the original function:

$$J^{**}(\hat{y}) = \text{conv}(J)(\hat{y}) \leq J(\hat{y}) \quad (14)$$

Since the analytical form of $J(\hat{y})$ is unknown, we construct the empirical convex hull $g(\hat{y})$ based on historical data points $\mathcal{D} = \{(y_i, c_i)\}_{i=1}^N$:

$$g(\hat{y}) = \max_{k=1, \dots, K} \{m_k \hat{y} + b_k\} \quad (15)$$

As the sample size $N \rightarrow \infty$, the empirical convex hull $g(\hat{y})$ converges to the theoretical Fenchel biconjugate $J^{**}(\hat{y})$, thereby capturing the global geometric structure of the scheduling cost C^{obj} .

However, theoretical optimality in the cost domain does not necessarily ensure physical feasibility in the time domain. Since the convex hull is constructed based on the marginal distribution of operating points, it treats each time step independently, thereby disregarding the temporal correlations inherent in wind power sequences. As a result, blindly minimizing the convex hull loss function $\mathcal{L}$ without incorporating physical regularization leads to mathematically trivial solutions. The following content provides a rigorous analysis of this solution degeneration (or mode collapse) phenomenon, highlighting the necessity of including the MSE term to enforce physical consistency.

Let the forecasting model f_θ parameterized by θ map input features x to a high-dimensional time-series vector $\hat{\mathbf{y}} \in \mathbb{R}^T$ over the scheduling horizon T :

$$\hat{\mathbf{y}} = f_\theta(x) = [\hat{y}_1, \hat{y}_2, \dots, \hat{y}_T]^T \quad (16)$$

The valid physical states of the wind power sequence lie on a low-dimensional manifold $\mathcal{M} \subset \mathbb{R}^T$ governed by temporal correlations constraints.

Consequently, the purely loss function $\mathcal{L}$ is structurally decomposed into T independent point-wise subproblems:

$$\mathcal{L}(\hat{\mathbf{y}}) = \sum_{t=1}^T g(\hat{y}_t) \quad (17)$$

The Hessian matrix of the loss $\mathcal{L}$ with respect to the output vector $\hat{\mathbf{y}}$ is strictly diagonal, implying zero curvature coupling between distinct time steps.

We examine the second-order partial derivatives:

$$H_{ij} = \frac{\partial^2 \mathcal{L}}{\partial \hat{y}_i \partial \hat{y}_j} \quad (18)$$

Since $\mathcal{L}$ is a separable sum of scalar functions, the cross-terms vanish for all $i \neq j$:

$$\frac{\partial^2 \mathcal{L}}{\partial \hat{y}_i \partial \hat{y}_j} = \frac{\partial}{\partial \hat{y}_j} g'(\hat{y}_i) = 0 \quad (19)$$

Thus, the Hessian matrix is given by:

$$\mathbf{H}_{\mathcal{L}} = \text{diag}(g''(\hat{y}_1), g''(\hat{y}_2), \dots, g''(\hat{y}_T)) \quad (20)$$

A diagonal Hessian confirms that the optimization landscape for $\hat{y}_t$ is mathematically independent of $\hat{y}_{t-1}$ and $\hat{y}_{t+1}$. The gradient signal backpropagated at time t contains no information regarding the temporal evolution of the sequence.

Since $g(p)$ is a convex function constructed from the lower envelope of costs, it possesses a global minimum set S^* . Typically, for a convex polytope, the minimum lies at a vertex v^* :

$$v^* = \arg \min_{p \in \text{Dom}(g)} g(p) \quad (21)$$

During training, the gradient vector $\nabla_{\hat{\mathbf{y}}} \mathcal{L}$ is given by the subgradients of the convex hull at each time step:

$$\nabla_{\hat{\mathbf{y}}} \mathcal{L} = [m_{k_1}, m_{k_2}, \dots, m_{k_T}]^T \quad (22)$$

where $m_{k_t} \in \partial g(\hat{y}_t)$. Since the optimization of each $\hat{y}_t$ is decoupled and driven by the same convex function $g(\cdot)$, the gradient descent dynamics will independently drive each component $\hat{y}_t$ toward v^* :

$$\lim_{\text{epoch} \rightarrow \infty} \hat{y}_t(x) = v^*, \quad \forall t \in \{1, \dots, T\}, \forall x \in \mathcal{X} \quad (23)$$

$$\lim_{\text{epoch} \rightarrow \infty} f_{\theta}(x) = \mathbf{v}^* = [v^*, v^*, \dots, v^*]^T \quad (24)$$

1.3 Our Fuction Effectiveness

In this final section, we rigorously prove the effectiveness of the proposed hybrid loss function. We demonstrate that the introduction of the convex hull regret term induces an Optimal Bias, which mathematically guarantees a reduction in expected scheduling costs compared to the unbiased estimator (pure MSE).

Let the total loss function $\mathcal{L}_{total}$ be defined as a weighted sum of the economic regret and the physical consistency term:

$$\mathcal{L}_{total}(\hat{y}) = \alpha \cdot [g(\hat{y}) - g(y)]_+ + \beta \cdot \|\hat{y} - y\|^2 \quad (25)$$

where $g(\cdot)$ is the convex hull surrogate of the scheduling cost, and $[z]_+ = \max(0, z)$ is the ReLU operator.

Consider the region where the regret term is active. The gradient of the total loss with respect to $\hat{y}$ is:

$$\nabla_{\hat{y}} \mathcal{L}_{total} = \alpha \cdot m_k + 2\beta(\hat{y} - y) \quad (26)$$

Setting the gradient to zero to find the stationary point (optimal prediction)

$$\alpha \cdot m_k + 2\beta(\hat{y}^* - y) = 0 \quad (27)$$

Solving for $\hat{y}^*$, we obtain the explicit form of the optimal prediction:

$$\hat{y}^* = y \quad \underbrace{-\frac{\alpha}{2\beta} m_k}_{\text{Optimal Bias Term } \delta^*} \quad (28)$$

Let ΔJ denote the change in scheduling cost when moving from the unbiased prediction y (MSE optimal) to the biased prediction $\hat{y}^*$ (Proposed optimal). Using the first-order Taylor expansion of the cost function around the ground truth y :

$$J(\hat{y}^*) \approx J(y) + \langle \nabla J(y), \hat{y}^* - y \rangle \quad (29)$$

Substituting the convex hull gradient m_k as the surrogate for $\nabla J(y)$ and substituting $(\hat{y}^* - y)$:

$$J(\hat{y}^*) \approx J(y) - \frac{\alpha}{2\beta} \|m_k\|^2 \quad (30)$$